

LLM Agents for Education: Advances and Applications

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Abstract

Large Language Model (LLM) agents are transforming education by automating complex pedagogical tasks and enhancing both teaching and learning processes. In this survey, we present a systematic review of recent advances in applying LLM agents to address key challenges in educational settings, such as feedback comment generation, curriculum design, etc. We analyze the technologies enabling these agents, including representative datasets, benchmarks, and algorithmic frameworks. Additionally, we highlight key challenges in deploying LLM agents in educational settings, including ethical issues, hallucination and overreliance, and integration with existing educational ecosystems. Beyond the core technical focus, we include in Appendix A a comprehensive overview of domain-specific educational agents, covering areas such as science learning, language learning, and professional development.

1 Introduction

Artificial intelligence techniques are increasingly used in education to enable personalized learning and intelligent tutoring (Chen et al., 2020; Zhai et al., 2021; Pedro et al., 2019; Yan et al., 2025b). While traditional educational data mining approaches (Shafiq et al., 2022; Khan and Ghosh, 2021; Abdelrahman et al., 2023; Song et al., 2022; Wang et al., 2022; Gao et al., 2021), such as knowledge tracing and cognitive diagnosis, have made significant progress in reshaping the human-learning paradigm by analyzing student behaviors and assessing knowledge states, they still face major challenges in real-world applications. These challenges include shallow contextual understanding, limited interactive capabilities, and difficulties in generating adaptive, personalized learning materials, etc (Zhu et al., 2024; Laak and Aru, 2024; Tan et al., 2023).

The strong natural language understanding of Large Language Models (LLMs) and the task automation capabilities of LLM agents make them valuable for addressing challenges in education (Weng, 2023; Wang et al., 2024a). First, memory enables LLM agents to retain both long-term knowledge about students’ study habits and short-term context from real-time interactions, enhancing contextual understanding and ensuring personalized learning experiences across various educational tasks (Zhang et al., 2024d). Second, tool use allows LLM agents to access external resources, perform complex calculations, and retrieve real-time information, enabling them to automate intricate educational tasks such as grading, knowledge retrieval, and adaptive content generation, thereby overcoming limited interactivity and enhancing engagement (Gao et al., 2023). Third, planning supports structured learning by decomposing complex topics, predicting optimal learning paths, and dynamically adjusting instructional strategies, allowing LLM agents to autonomously guide students through personalized learning experiences (Huang et al., 2024b).

In addition to these core architectural capabilities, we identify personalization, explainability, and multi-agent communication as essential for effective educational LLM agents, as they enable nuanced instructional behavior and collaborative reasoning. Personalization allows agents to tailor instruction to individual learners’ needs and preferences (Razafinirina et al., 2024). Explainability enables them to provide interpretable justifications for feedback and decisions (Abu-Rasheed et al., 2024b). Multi-agent communication facilitates role-based collaboration, such as between a planner and a critic, to improve robustness and task coverage (Wu et al., 2023a). By integrating these features, LLM agents enhance understanding and engagement while streamlining educational workflows for greater adaptability and efficiency.

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In this survey, we provide a comprehensive review of LLM agents in educational settings, with a focus on their underlying technical foundations and general-purpose pedagogical capabilities. We begin by introducing the core abilities of educational LLM agents, including memory, tool use, planning, personalization, explainability, and multi-agent communication, and discuss their potential to automate and enhance diverse educational tasks. Given the highly applied nature of the education domain, we propose a task-centric taxonomy in Figure 1 that organizes recent advances around core educational tasks. We categorize educational LLM agents based on their roles in supporting teachers and students, capturing both instructional and learning-focused functions: (1) Teaching Assistance Agents, which support educators by automating tasks such as *Classroom Simulation (CS)*, *Feedback Comment Generation (FCG)*, and *Curriculum Design (CD)*; and (2) Student Support Agents, which facilitate personalized learning through *Adaptive Learning (AL)*, *Knowledge Tracing (KT)*, and *Error Correction and Detection (ECD)*. The overview of LLM agents for education and illustrative examples of each task are presented in Figure 2. We further highlight critical challenges in deploying these agents, including ethical issues, hallucination and overreliance, and integration into existing educational ecosystems. Finally, we compile essential datasets, benchmarks, and evaluation protocols to support future research on LLM agent-based educational systems. We summarize our contributions as follows:

- **Novel task-centric taxonomy.** We propose a structured taxonomy that categorizes LLM agents based on core educational tasks, highlighting roles in teaching assistance and student support to unify analysis across applications.
- **Current challenges and future directions.** We analyze critical challenges that need to be addressed for the effective deployment of LLM agents for education, including issues related to ethical issues, hallucination, and integration into real-world educational ecosystems.
- **Compilation of essential resources.** We compile comprehensive datasets and benchmarks to support future research efforts and facilitate the development of more robust and effective LLM-driven educational solutions.¹

¹Due to the page limit, we present more details in Ap-

Beyond the main technical focus, Appendix A provides an overview of domain-specific educational agents, including applications in science learning, language learning, and professional development. We outline domain-specific challenges and review recent advances, benchmarks, and datasets. Readers are encouraged to refer to the appendix for further details.

2 LLM Agents for Education

LLM agents have a set of connected abilities that help them handle complex tasks and provide meaningful support in education. These core features work together to let LLM agents do more than just find information—they can interact in ways that are flexible, responsive, and tailored to each learner. Their main strengths in education include strong memory, the ability to use tools, planning skills, personalization, clear explanations, and the ability to work with other agents. Specifically,

Memory. Memory in LLM agents includes long-term (foundational knowledge, e.g., commonsense) and short-term (current interaction data) components (Sumers et al.). Active management via summarization and retrieval ensures relevant context is maintained (Chen et al., 2023; Liang et al., 2023; Zhong et al., 2023). This allows agents to track student progress and personalize responses, though sophisticated filtering is crucial to maintain the quality against noisy interaction data.

Tool Use. To overcome limitations like knowledge cutoffs or calculation difficulties, LLM agents use external tools such as search engines, databases, or APIs (Qin et al., 2023; Zhuang et al., 2023). This expands their functionality, providing access to current information and specialized capabilities. By integrating these tools, LLM agents expand their functional capabilities, ensuring that they can provide accurate and relevant information while supporting diverse educational tasks.

Planning. Planning allows agents to actively support learning by breaking down complex goals into smaller, manageable steps and adjusting based on how the student is doing (Yao et al., 2023; Valmeekam et al., 2023). This means understanding the learning goals, creating step-by-step plans, personalizing the learning path, and making changes based on feedback and progress. For longer-term tasks, LLMs can act as “meta-controllers,” using methods like “Pedagogical

pendix B

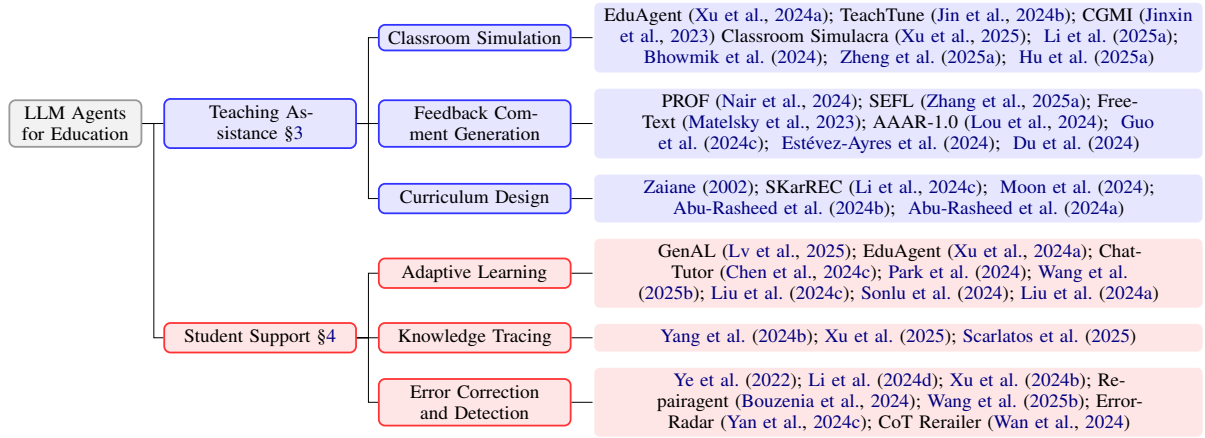


Figure 1: Taxonomy of representative research on LLM agents for education.

Steering” to stay aligned with teaching goals, which thus leads to dynamic, emergent curricula co-created with students (Zhang et al., 2025c).

Personalization. Personalization is a hallmark of effective education, and LLM agents excel in this area by adapting to individual learning styles and needs (Chen et al.). They can serve as tutors, teaching assistants, or even simulate peer interactions to enhance the learning experience (Jin et al., 2024b; Guo et al., 2024c). Additionally, LLM agents can simulate user behaviors in recommendation systems, which can be applied to personalize educational content (Li et al., 2024c; Chu et al., 2025). Such a technology of personalization transforms traditional education, making it more accessible and tailored to each student’s unique needs.

Explainability. Explainability is crucial for building trust and facilitating learning in educational settings (Wu et al., 2024b). LLM agents must provide clear, step-by-step explanations that are understandable to students. Clear explanations are particularly important in subjects like STEM, where step-by-step reasoning is essential for student understanding (Nair et al., 2024; Zhang et al., 2025a). Consequently, the focused effort required to make educational LLM agents more transparent and their decision-making processes more scrutable.

Multi-Agent Communication. LLM agents can facilitate richer interactions in collaborative learning environments (Chen et al.; Wu et al., 2023a). Multiple agents could coordinate on group projects or simulate peer learning, fostering critical thinking and diverse perspectives (Jinxin et al., 2023; Yue et al., 2024). This approach could support complex simulations of teamwork and real-world professional settings, especially when guided by new teaching frameworks for organization and eval-

uation (Xu et al., 2024a).

Table 1 summarizes the core capabilities required by each educational task supported by LLM agents. For each task, we highlight the primary capabilities identified from the reviewed literature. However, certain studies have also explored additional capabilities beyond these primary ones to enhance specific tasks—for example, Guo et al. (2024c) leverage multi-agent communication to improve feedback correctness. We omit these secondary capabilities from the table to more clearly illustrate the primary focus areas associated with each task.

3 Agent for Teaching Assistance

Agents for teaching assistance are designed to support educators in the learning environment. These agents leverage LLMs to provide personalized, scalable, and efficient support across various aspects of the educational process. Their primary objectives are to enhance teaching quality, enrich student learning experiences, and reduce educators’ workload. By incorporating advanced capabilities, LLM agents can effectively support key educational tasks, including classroom simulation (§3.1), feedback comment generation (§3.2), and curriculum design (§3.3).

3.1 Classroom Simulation

Classroom simulation refers to the ability of teaching agents to replicate and model various classroom scenarios, such as student-teacher dialogues, collaborative learning activities, and problem-solving tasks. These simulations create dynamic and interactive learning environments where educators can experiment with different teaching strategies, assess student reactions, and receive real-time feed-

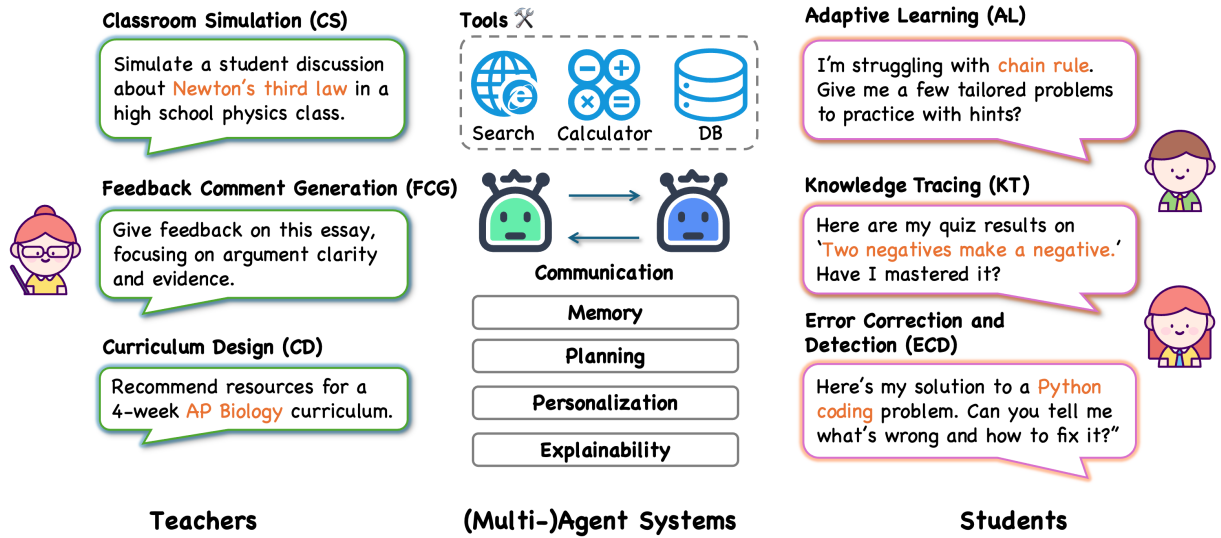


Figure 2: The overview of LLM Agents for education. Teachers and students interact with LLM agents by submitting task-specific prompts. The agents respond using core capabilities such as memory, planning, tool use, and personalization to carry out tasks that support instruction and learning.

Task	Memory	Tool Use	Planning	Personalization	Explainability	Multi-Agent Comm.
CS	✓	—	✓	—	—	✓
FCG	—	—	—	✓	✓	—
CD	✓	✓	✓	✓	—	—
AL	✓	—	✓	✓	—	—
KT	✓	—	—	✓	—	—
ECD	—	✓	—	✓	✓	—

Table 1: Mapping of educational tasks to essential LLM agent capabilities. ✓ indicates a primary capability; others may also be used as supporting components in some systems.

back on how various pedagogical approaches may unfold. By simulating these classroom dynamics, educators can refine their methods, anticipate student challenges, and enhance overall instructional effectiveness, all without the constraints of a physical classroom setting.

As shown in Table 1, effective classroom simulation relies on memory, planning, and multi-agent communication to accurately model student behavior. Previous studies (Xu et al., 2024a; Jin et al., 2024b; Li et al., 2025a) demonstrate that LLM-based agents can predict fine-grained student behaviors across diverse personas and past learning patterns, aligning closely with real teachers' expectations. To enhance simulation, the CGMI framework (Jinxin et al., 2023) uses a tree-based cognitive architecture with memory, reflection, and planning modules to simulate roles like teacher, student, and supervisor, improving realism. Similarly, Classroom Simulacra (Xu et al., 2025) incorporates

a transferable iterative reflection module for more accurate behavior simulation. These systems enable automated interactions that reduce educators' task loads while broadening the exploration of student profiles. Simulations can also test educational strategies tailored to different profiles, enhancing teaching quality, as shown in studies by Bhowmik et al. (2024) and Zheng et al. (2025a). Additionally, Hu et al. (2025a) demonstrate how LLMs can refine teaching plans through integrated simulations. To sum up, classroom simulations can be leveraged to test different educational strategies tailored to diverse student profiles, ultimately enhancing the quality of education.

3.2 Feedback Comment Generation

Providing timely, relevant, and constructive feedback is a cornerstone of effective education. Teaching assistance agents can generate automated feedback comments on students' assignments, quizzes,

and projects.

For example, Guo et al. (2024c) leverage multi-agent communication to provide accurate feedback to students through a two-agent system. Specifically, Agent 1 generates initial feedback based on the students' responses, while Agent 2 evaluates and refines this feedback to prevent overpraise and excessive inferences. Furthermore, Nair et al. (2024) design a training strategy called PROF, which trains an automated LLM-based writing comment generator through reinforcement learning. This system adopts an iterative pipeline to simulate various student writing styles and incorporates a more advanced revision model (e.g., GPT-4) to provide the quality of the feedback as rewards. Similarly, SEFL (Zhang et al., 2025a) enhances feedback generation by having LLM agents role-play both students and teachers to generate data, which is then used to fine-tune models and improve feedback capabilities. These systems have also been deployed in real-world applications, such as FreeText (Matelsky et al., 2023), which pairs student responses with teacher-provided criteria, enabling the agent to identify strengths and weaknesses and provide targeted feedback for improvement. Beyond traditional feedback, advanced LLM agents are now capable of handling more complex, expertise-intensive tasks. For example, Du et al. (2024) explores the potential of LLM agents as assistants for natural language processing paper reviewing tasks, while AAAR-1.0 (Lou et al., 2024) evaluates agents' capabilities in areas such as equation inference, experiment design, paper weakness analysis, and review critique, revealing their potential in conducting research tasks.

However, another line of research highlights that agent-generated feedback still faces challenges in handling complex tasks, such as programming and the review of professional academic papers (Estévez-Ayres et al., 2024; Lou et al., 2024). For instance, these agents may struggle with concepts like starvation and deadlocks, leading to inaccurate or incorrect feedback. Future work could focus on integrating external tools (e.g., search engines) and enhancing memory mechanisms to better address complex problem-solving scenarios, as well as refining the personalization of feedback across diverse learning contexts.

3.3 Curriculum Design

To ensure that students follow personalized learning paths aligned with their knowledge level and

domain, it is crucial to develop effective curriculum design strategies. This is a complex task that relies on multiple LLM agent capabilities, including memory, tool use, planning, personalization, and explainability.

Zaiane (2002) first introduces recommendation systems into e-learning, utilizing web mining techniques to suggest online learning activities or shortcuts on course websites. The integration of LLM-based agents has recently enabled more sophisticated curriculum design through dynamic sequencing and content adaptation. Curricula can be built using either retrieval-based or generation-based methods. Retrieval-based methods involve agents accessing a database or their own memory to suggest existing resources—such as textbooks, research papers, or online content—based on student queries, past behavior, or content similarities (Shahzad et al., 2025; Li et al., 2024c). In contrast, generative methods create new learning content tailored to an individual student's learning style, knowledge gaps, and interests (Moon et al., 2024). Moreover, to enhance the understanding and acceptance of recommended content, these agents need to provide reasons for their recommendations. Explaining the rationale behind suggestions fosters trust and enables students to make more informed decisions about the resources they engage with. For example, Abu-Rasheed et al. (2024a,b) incorporate knowledge graphs to guide curriculum pathways toward curated and trustworthy sources. This approach not only enhances the interpretability of recommendations but also reduces the risk of generating misinformation, thereby improving the quality of the learning experience.

Looking ahead, future work should focus on integrating agents with adaptive learning systems to dynamically adjust recommendations in response to real-time student performance. In addition, a hybrid approach that combines generative and retrieval-based methods may improve both the accuracy and diversity of curriculum content. Also, incorporating multimodal resources such as interactive media, video, and immersive simulations could further enhance the learning experience.

4 Agent for Student Support

LLM agent-based student support systems aim to provide real-time personalized assistance without requiring direct teacher involvement. Unlike traditional rule-based systems, these agents offer inter-

active and adaptive feedback, enabling students to progress at their own pace. Core educational tasks in student support include adaptive learning (§4.1), knowledge tracing (§4.2), and error correction and detection (§4.3).

4.1 Adaptive Learning

LLM agents offer the potential to build self-sustaining adaptive learning systems that operate without direct teacher involvement. These systems dynamically tailor instruction based on student performance, enabling personalized learning at scale. Building on the abilities of memory and personalization, agents are able to maintain and update a structured representation of the learner, commonly referred to as a student profile. This profile informs content selection, pacing, and feedback strategies.

Several implementations exemplify this adaptive approach. GenAL (Lv et al., 2025) integrates external tools such as automated programs and web searches to construct comprehensive student profiles and inform instructional planning. Based on these profiles, the agent assigns tasks that align with the student’s current knowledge level and updates its memory dynamically. EduAgent (Xu et al., 2024a) introduces a structured profiling mechanism comprising four distinct cognitive patterns: gaze behavior linked to physiological memory, motor behavior mapped to motor memory, cognitive state associated with cognitive memory, and post-course assessments contributing to knowledge memory. This structured representation enhances adaptive decision-making by providing a multi-faceted view of student learning states. Chen et al. (2024c) propose a system consisting of interaction, reflection, and reaction, with each component composed of specific LLM tools and memory modules. Furthermore, a meta-agent is introduced to control the information flow through these agents.

Some recent research focuses on modeling students through modular memory or state components that capture *cognitive*, *affective*, and *psychological* dimensions. These representations are often formalized within a state–action framework, where the state space encodes learner traits and the action space governs instructional adaptations. A typical cognitive state representation includes tracking a student’s knowledge proficiency, comprehension levels, and misconceptions (Park et al., 2024; Liu et al., 2024c,a), allowing the agent to tailor explanations, adjust difficulty levels, and reinforce concepts dynamically. Recent studies high-

light the importance of affective state modeling, as emotional factors such as motivation, interest, and self-efficacy significantly influence learning outcomes. For instance, Park et al. (2024) propose an affective state model that enables agents to adjust feedback tone, provide encouragement, and regulate pacing to maintain engagement. Another crucial dimension of adaptation involves learning preferences and personality traits. These studies integrate personality with memory design, tracking the psychological state of the students. Wang et al. (2025b) integrate learning preferences into state modeling, recognizing that students process information differently depending on instructional format and modality. Adapting content to these preferences enhances retention and learning efficiency. Moreover, Liu et al. (2024c) apply the Big Five personality model (Roccas et al., 2002) to personalize tutoring strategies, acknowledging that individual differences shape learning experiences (Sonlu et al., 2024).

Emerging works also explore multi-agent systems for adaptive learning, where specialized agents collaborate to enhance personalization. For example, Wang et al. (2025b) design five agents—Gap Identifier, Learner Profiler, Dynamic Learner Simulator, Learning Path Scheduler, and Content Creator—to deliver goal-oriented, personalized instruction. Similarly, OATutor (Pardos et al., 2023) provides an experimental platform for modular adaptive learning, allowing researchers to design scalable, domain-general tutoring agents.

4.2 Knowledge Tracing

Knowledge tracing is essential for monitoring a learner’s evolving understanding and predicting future performance. While traditional methods use statistical or deep learning models to estimate mastery, LLM agents offer a more dynamic, personalized approach by leveraging natural language understanding and adaptive instruction.

Recent advancements have explored multi-agent frameworks for knowledge tracing. For instance, Yang et al. (Yang et al., 2024b) propose a multi-agent system with three specialized agent roles: administrator, judge, and critic. In this framework, the administrator delegates knowledge tracing tasks to judges, who collaborate through discussions to assess the student’s cognitive state. The critic agent then evaluates the outcome and determines whether the assessment criteria are met, ensuring a structured yet flexible knowledge tracing process.

Other agent-based approaches explore alternative strategies for modeling student knowledge. [Xu et al. \(2024a\)](#) propose simulating students as different personas, allowing agents to adaptively trace knowledge progression based on varied learning profiles. Meanwhile, [Scarlato et al. \(2025\)](#) employ dialogue-driven interactions to probe students' conceptual boundaries, using conversational exchanges to refine knowledge estimation dynamically.

4.3 Error Correction and Detection

Error detection and correction help students refine their understanding through real-time, context-aware feedback. LLM agents can identify mistakes across domains such as writing, programming, and math, and adapt feedback to the learner's proficiency, acting as intelligent reviewers and assistants ([Ye et al., 2022](#); [Li et al., 2024d](#)).

Agent-based systems leverage state representations and adaptive inference mechanisms to track error patterns and misconceptions dynamically ([Park et al., 2024](#); [Liu et al., 2024c](#); [Wang et al., 2025b](#); [Bouzenia et al., 2024](#)). Recent advancements extend this capability into the multi-modal domain, incorporating direct analysis of student-generated drafts. [Xu et al. \(2024b\)](#) propose a multi-modal LLM framework that processes handwritten or digitally drafted student work. The system first extracts and converts draft content into natural language, enabling the agent to interpret and analyze handwritten responses. The agent then provides indirect yet effective instructional feedback, guiding students toward self-correction and deeper comprehension. Moreover, CoT Rerailer ([Wan et al., 2024](#)) designs a derailment identification process and a rerailment process to conduct error detection when solving math questions. [Zhang et al. \(2025d\)](#) propose the **MathCCS benchmark** and introduce a **sequence error analysis** framework that leverages multi-agent collaboration. As the first benchmark for multimodal error detection, ErrorRadar ([Yan et al., 2024c](#)) provides a valuable data foundation for developing multimodal agents in this task.

5 Challenges and Future Directions

In this section, we discuss key challenges that must be addressed to ensure the effective, reliable, and ethical deployment of LLM agents in educational settings. We focus on three critical areas: (1) privacy, bias, and fairness; (2) hallucination and overreliance; and (3) integration with existing educa-

tional ecosystems. For each of these challenges, we outline potential research directions aimed at improving the robustness, trustworthiness, and practical applicability of LLM agents in real-world education environments.

5.1 Privacy, Bias and Fairness

Analysis. LLM agents process vast datasets, often containing sensitive personal information, leading to potential privacy risks. Studies highlight low technological readiness and insufficient privacy measures in educational contexts ([Yan et al., 2024a](#)). Emerging research ([He et al., 2024a](#); [Gan et al., 2024](#); [Zhang et al., 2024c](#); [Hua et al., 2024](#); [Huo et al., 2025](#); [Chen et al., 2025b](#)) underscores new privacy and security concerns, emphasizing the need for stronger data protection mechanisms ([Huang, 2023](#); [Ismail, 2025](#); [Khan, 2024](#)). Additionally, bias in LLMs remains a pressing concern, as models trained on large datasets can inadvertently reinforce stereotypes and disparities, affecting educational fairness. Recent work calls for bias mitigation strategies to promote equitable learning experiences ([Adewumi et al., 2024](#); [Aird et al., 2024](#); [Mehrotra et al., 2024](#)). Addressing these biases is essential to ensuring inclusive, unbiased educational outcomes.

Directions. To overcome the above issues, a number of future directions can be explored: (i) *Unlearning for privacy preservation*: leverage advances in machine unlearning ([Liu et al., 2025](#)) to enable agents to retain useful knowledge while selectively forgetting sensitive user data when required. (ii) *Bias detection and mitigation*: develop automated fairness-checking models that evaluate real-time content generated by LLM agents to detect biased explanations, language, or examples. (iii) *Culturally adaptive LLM Agents for global education*: train multilingual, culturally aware educational agents that dynamically adjust explanations based on regional learning norms, historical perspectives, and diverse curricula.

5.2 Hallucination and Overreliance

Analysis. The “hallucination” phenomenon, in which LLMs generate plausible but incorrect or nonsensical information, poses a significant challenge to their reliability in educational contexts ([Zhang et al., 2023](#)). Such inaccuracies can mislead learners by presenting false information with confident and authoritative language, making errors difficult to detect and potentially leading to

misconceptions (da Silva et al., 2024; Jho, 2024; Ho et al., 2024). For example, AI-generated content may fabricate historical events or scientific facts that students unknowingly accept as true. This risk is further amplified by overreliance, as students and educators may accept AI-generated responses without sufficient critical evaluation. Research has shown that excessive dependence on AI systems can hinder skill acquisition and reduce meaningful engagement with learning materials (Milano et al., 2023; Krupp et al., 2024; Adewumi et al., 2023).

Directions. Some directions can be explored to mitigate hallucinations in LLM agents for education: (i) *Self-correcting AI tutors*: develop LLM agents with self-reflection capabilities (Renze and Guven, 2024), where models review, verify, and refine their own generated content before presenting it to students. (ii) *Hybrid Human-AI feedback loops for educational content verification*: develop teacher-in-the-loop AI systems where educators can review and correct AI-generated responses, refining agents performance over time. (iii) *Pedagogical-aware educational agents*: design agentic frameworks aligned with human pedagogical expertise to mitigate overreliance on AI-generated content.

5.3 Integration with Existing Educational Ecosystems

Analysis. Although LLM agents hold great potential for automating educational practices, it is essential to consider how they can be effectively integrated into existing human-centered educational paradigms. One major challenge is the lack of structured frameworks for integrating LLM agents into educational systems. While models like the FOKE framework (Hu and Wang, 2024) combine foundation models, knowledge graphs, and prompt engineering to provide interactive and explainable learning services, broader adoption requires scalable models that can be validated in diverse real-world educational settings. Additionally, LLMs have been explored as tools to enhance creativity and collaboration in project-based learning (PBL), supporting students through brainstorming, problem-solving, and project execution. However, studies indicate that their effectiveness is limited by the absence of structured guidance frameworks that help educators and students seamlessly incorporate LLM agents into PBL workflows (Zha et al., 2024). Another critical challenge is ensuring equitable access to LLM-powered educational tools, particularly in underfunded schools and institu-

tions with limited AI infrastructure. Platforms such as AI-VERDE (Mithun et al., 2025) aim to democratize access by providing a unified *LLM-as-a-platform* service with built-in access control, privacy-preserving mechanisms, and budget management. However, achieving widespread adoption still depends on scalable and cost-effective deployment strategies that can support educational institutions at different resource levels.

Directions. Future research should focus on developing standardized frameworks to guide the structured deployment of LLM agents in personalized learning, PBL, and assessment. For example, expanding models like FOKE with adaptive learning strategies, multimodal content, and real-time feedback could enhance instructional effectiveness. Additionally, integrating interactive AI tutors that support student collaboration, project tracking, and contextual guidance would further improve PBL applications. To promote equitable access, develop cost-effective AI tutors through cloud-based and decentralized models would make LLM-powered learning tools more accessible to a wider range of institutions. Finally, to support meaningful integration of LLM agents into educational ecosystems, future work should move beyond task accuracy and explore more practical metrics such as learning gains, user trust, and engagement, which calls for the development of novel education-oriented benchmarks and datasets.

5.4 Multimodal LLM Agents for Education

Analysis. Despite rapid progress (Bewersdorff et al., 2025; Küchemann et al., 2025), deploying multimodal LLM agents in education faces several unresolved challenges. First, integration of heterogeneous modalities such as speech, images, videos, and VR simulations is technically demanding. Models often overfit to text while underutilizing other modalities (Zheng et al., 2025b). Second, cultural and contextual alignment is still limited: current agents may misinterpret classroom discourse, gestures, or culturally grounded communication patterns, reducing the accuracy and inclusiveness of feedback. Third, real-time adaptability is critical yet underexplored. For example, VR-based or speech-driven learning environments demand low-latency responses, but existing agents struggle with efficiency when reasoning across modalities (Voultsiou and Moussiades, 2025).

Directions. Some directions can be explored to advance the use of multimodal LLM agents in

education: (i) *Robust multimodal fusion*: leverage contrastive pretraining and modality-specific adapters to balance contributions from text, speech, and vision, reducing overfitting to dominant modalities (Nguyen et al., 2022). (ii) *Context-aware personalization*: incorporate localized knowledge graphs and human-in-the-loop feedback to better align agent reasoning with diverse classroom discourse and cultural norms. (iii) *Latency-aware optimization*: deploy lightweight distilled models and selective modality routing to ensure real-time responsiveness in VR and speech-driven learning environments.

6 Conclusion

In this survey, we presented a comprehensive review of LLM agents for education, focusing on their technical foundations and their potential to transform personalized learning, intelligent tutoring, and pedagogical automation. We proposed a task-centric taxonomy that categorizes LLM agents into Teaching Assistance and Student Support, highlighting their core capabilities such as memory augmentation, tool use, planning, and personalization. We also examined key research challenges, including ethical issues, hallucination and over-reliance, and integration with existing educational ecosystems, which must be addressed to ensure reliable and ethical deployment. To support continued progress in this field, we compiled critical datasets, benchmarks, and evaluation methodologies. As LLM agents continue to evolve, their influence on education will expand. Realizing their full potential, however, will require both technical rigor and thoughtful system design. We hope this survey provides a solid foundation for future research and the development of AI-powered educational systems.

Limitations

Considering the rapid development of LLM agents for education, it is possible that some of the most recent advancements may not have been captured at the time of writing. Nevertheless, we have made every effort to ensure that all foundational and representative works are included to provide a comprehensive and accurate overview of the field. Specifically, we searched Google Scholar using keywords like “LLM agents for education”, “LLM agents for classroom simulation”, “LLM agents for adaptive learning”, and “LLM agents for knowledge tracing” covering publications from 2020 to May 2025.

We included studies focusing on LLM agent-based methods but excluded those solely on LLM-based approaches.

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A Domain-Specific Educational Agents

Recent research on LLM agents in education has also shown growing interest in domain-specific applications. We explore their use in *science learning*, *language learning*, and *professional development*, focusing on their algorithmic frameworks, agentic designs, and relevant datasets and benchmarks. In Figure 3, we present a systematic taxonomy of domain-specific educational agents.

A.1 Agent for Science Learning

An agent for science learning is an intelligent system powered by LLMs, designed to assist students in acquiring and applying scientific knowledge through personalized, interactive experiences (Yan et al., 2025b; Raihan et al., 2025; Ng et al., 2024; Brown and Cruz Castro, 2025). The significance of these agents in education lies in their ability to offer tailored feedback, enhance conceptual understanding, and promote active engagement with complex scientific ideas. In the following sections, we explore the impact of LLM agents in four key scientific disciplines: *mathematics* (§A.1.1), *physics* (§A.1.2), *chemistry* (§A.1.3), and *biology* (§A.1.4), as well as their broader contributions to *general scientific discovery* (§A.1.5).

A.1.1 Mathematics

In mathematics, LLM agents provide substantial support by helping students navigate complex problems and reinforcing their understanding of abstract concepts (Yan et al., 2024b; Xiong et al., 2024; Swan et al., 2023; Yan et al., 2024c; Wu, 2025; Mitra et al., 2024). For instance, Gou et al. (2023) introduce TORA (Tool-integrated Reasoning Agents), a framework that integrates natural language reasoning and program-based tool use to handle mathematical reasoning. MathAgent (Yan et al., 2025a) similarly proposes Mixture-of-Math-Agent framework to address multimodal error detection in real-world K-12 scenarios, and flexibly transform the visual information of different types of questions into forms that are more easily understood by LLMs (e.g., converting plane geometry images into formalized expression). Additionally, MathChat (Wu et al., 2023b) serves as a conversational mathematical problem-solving agent, which consists of a chat-based LLM agent and a tool-based user agent. Furthermore, Xiong et al. (2024) propose to use reinforcement learning from human feedback (RLHF) to further improve tool-integrated

agents for mathematical problem-solving, and formulate this method as a Markov decision process, distinguishing it from the typical contextual bandit approach used in RLHF. Besides, MACM (Lei et al., 2025) discuss the limitations of LLMs in handling complex mathematical logical deduction, thus introducing a multi-agent system, which comprises three interactive agents: Thinker, Judge, and Executor.

A.1.2 Physics

In the field of physics, LLM agents help students make sense of challenging concepts and offer interactive tools to simulate physical phenomena (Pang et al., 2024; Mower and Bou-Ammar, 2025; Barman et al., 2025; Feng et al.; Jiang and Jiang, 2024; Yan and Lee, 2024). Wang et al. (2023) introduce NEWTON, the first pipeline and benchmark to explore the physical reasoning abilities of LLMs. Furthermore, Kortemeyer (2023) describe a case study exploring if an LLM agent can pass an introductory calculus-based physics course. In addition, Physics Reasoner (Pang et al., 2024), a novel knowledge-augmented framework for physics problem-solving, leverages a comprehensive formula set and detailed checklists to ensure accuracy and completeness. It can serve as an agent consisting of three stages - problem analysis, formula retrieval, and guided reasoning. Besides, Ma et al. (2024a) describe the Scientific Generative Agent (SGA), a bilevel optimization framework designed for physical scientific discovery, and highlight the use of LLMs for generating and revising scientific hypotheses and implementing an exploit-and-explore strategy.

A.1.3 Chemistry

Chemistry education also benefits greatly from LLM agents, which can explain molecular structures, chemical reactions, and experimental processes in an engaging and interactive way (Ramos et al., 2025; M. Bran et al., 2024; Yu et al., 2024; Guo et al., 2023; Tsai et al., 2023). For example, ChemCrow (M. Bran et al., 2024) is the first LLM chemistry agent capable of autonomous planning and execution of chemical syntheses, including an insect repellent and three organocatalysts. Yu et al. (2024) further present ChemAgent, an enhanced chemistry agent improved over ChemCrow, with a focus on two essential cognitive abilities of chemistry problem-solving: reasoning and grounding. Besides, Curie (Kon et al., 2025) is an agent framework aimed at incorporating rigor into the experi-

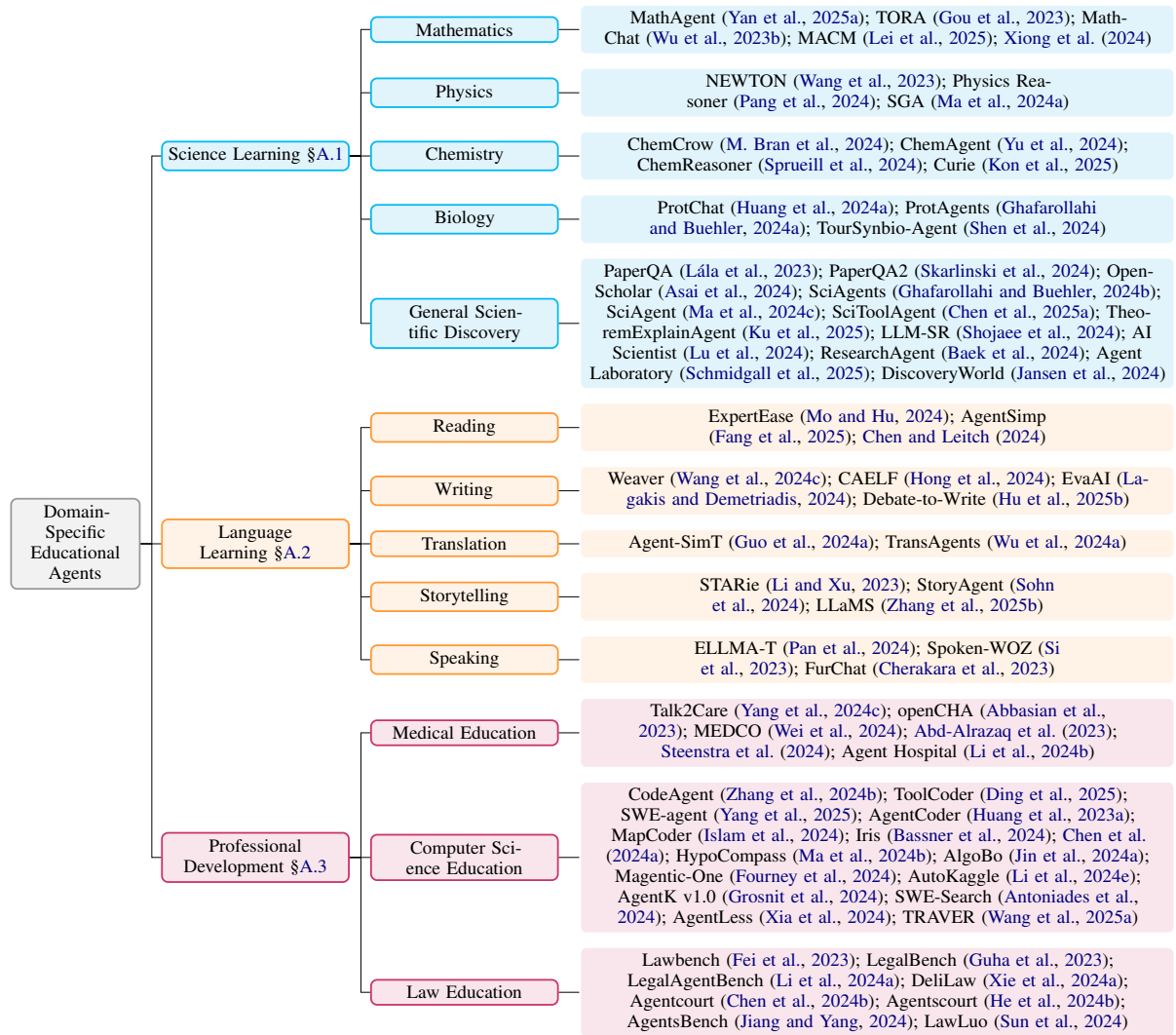


Figure 3: Taxonomy of domain-specific educational agents.

mentation process via three core elements: an intra-agent rigor module to boost reliability, an inter-agent rigor module to ensure systematic control, and an experiment knowledge module to improve interoperability. Recent studies have explored the capabilities of LLMs in complex chemical discovery (Yang et al., 2024d; Ruan et al., 2024; Sprueill et al., 2024; Moret et al., 2023; Jablonka et al., 2024), and their potential can be advanced by leveraging the interactivity of agent-based tool use and the flexibility of planning strategies (Song et al., 2024; Ramos et al., 2025).

A.1.4 Biology

In biology, LLM agents enhance learning by offering detailed explanations of biological processes and providing interactive experiences to explore living systems (Yan et al., 2025b; Bhattacharya et al., 2024; Sripathi et al., 2024; Zhao et al., 2025; Gao et al., 2024). For example, ProtChat (Huang et al., 2024a) is a multi-agent tool leveraging GPT-4 and Protein Language Models for seamless protein analysis automation, thus evolutionizing the complexities of protein sequence interpretation. ProtAgents (Ghafarollahi and Buehler, 2024a) is introduced as a multi-agent modeling framework that combines state-of-the-art LLMs with diverse tools to tackle protein design and analysis. It consists of a team of agents: User, Planner, Assistant, Critic, and Group Chat Manager. Besides, Shen et al. (2024) present TourSynbio-Agent, an innovative agent framework that leverages TourSynbio-7B’s protein understanding ability to perform various protein engineering tasks, such as mutation analysis, inverse folding, and visualization.

A.1.5 General Scientific Discovery

LLM agents support general scientific discovery by assisting students in data interpretation, hypothesis testing, and creative problem-solving (Yan et al., 2025b; Chen et al., 2024d; Ghafarollahi and Buehler, 2024b; Chen et al., 2025a; Schmidgall et al., 2025). These LLM agents, such as PaperQA (Lála et al., 2023), PaperQA2 (Skarlinski et al., 2024), OpenScholar (Asai et al., 2024), SciAgents (Ghafarollahi and Buehler, 2024b), TheoremExplainAgent (Ku et al., 2025), and LLM-SR (Shojaee et al., 2024), can analyze complex scientific datasets, helping students uncover patterns and trends that may not be immediately apparent. In addition, Narayanan et al. (2024) present Aviary, an extensible gymnasium for language agents for

three challenging scientific tasks: manipulating DNA constructs for molecular cloning, answering research questions by accessing scientific literature, and engineering protein stability. Furthermore, both SciAgent (Ma et al., 2024c) and SciToolAgent (Chen et al., 2025a) extend to a tool-augmented scientific reasoning setting with the help of domain-specific tools. Besides, Agent Laboratory (Schmidgall et al., 2025) emerges as an agent framework that automates the research process of three phases (Literature Review, Experimentation, and Report Writing) via various LLM agents (PhD, Postdoc, ML Engineer, *etc.*).

A.2 Agent for Language Learning

The integration of LLM agents into language learning is revolutionizing how core competencies—reading, writing, listening, and speaking—are taught and practiced (Ye et al., 2025b). These skills form the foundation of effective communication and language acquisition, and recent advancements in LLM-based agents have significantly enhanced how learners interact with and acquire these skills. Below, we introduce recent advancements in each subdomain, highlighting the role of LLM agents in enhancing pedagogical outcomes in language acquisition for students and second language (L2) speakers through engaging and adaptive approaches (Huang et al., 2023b; Ye et al., 2023b,a).

A.2.1 Reading

Reading comprehension is a vital component of language learning, and LLM agents are playing an increasingly important role in enhancing students’ reading abilities. For instance, ExpertEase (Mo and Hu, 2024) employs a multi-agent framework to adapt documents for grade-specific audiences, simulating expert-teacher-student collaboration to enhance comprehension. AgentSimp (Fang et al., 2025) tackles document-level simplification by leveraging multiple agents with distinct roles to ensure coherence and accessibility. Additionally, various LLMs (Adetayo et al., 2024) have been used as academic reading companions, demonstrating improved engagement and understanding of complex qualitative texts in educational settings (Chen and Leitch, 2024).

A.2.2 Writing

The development of writing skills has benefited significantly from NLP tasks like explainable gram-

mathematical error correction (EXGEC) (Ye et al., 2024b, 2025a; Zou et al., 2025) and automatic essay scoring (AES) systems (Su et al., 2025). Weaver (Wang et al., 2024c), a family of LLMs fine-tuned for writing tasks (Ye et al., 2023c, 2024c), outperforms generalist LLMs like GPT-4 in generating human-like narratives. Moreover, Weaver natively supports retrieval-augmented generation (RAG) and function calling, serving as a qualified foundational model for LLM agents. For interactive feedback on student essays, CAELF (Hong et al., 2024) introduces a multi-agent framework that enables interactive essay feedback. By combining Teaching-Assistant agents' evaluations with teacher-agent arbitration, students can contest grades and engage with the feedback, addressing the "black box" limitations of traditional automated scoring. Inspired by the process of human debate, Debate-to-Write (Hu et al., 2025b) constructs a persona-based multi-agent framework that can enable agents to collaboratively debate, discuss ideas, and form a comprehensive plan for argument writing.

A.2.3 Translation

LLM agents demonstrate remarkable advancements in both simultaneous (Koshkin et al., 2024; Guo et al., 2024b) and literary translation (Cheng et al., 2024; Škobo and Petričević, 2023). Translation tasks benefit from LLM agents through their ability to integrate specialized tools and orchestrate multi-agent collaboration. Agent-SiMT (Guo et al., 2024a) combines the decision-making capabilities of a Simultaneous Machine Translation (SiMT) policy agent with the generative power of a translation agent, achieving state-of-the-art performance in simultaneous translation by dynamically balancing reading and generation actions. For literary translation, TransAgents (Wu et al., 2024a) employs a multi-agent framework to replicate the complex workflows of human translation teams, addressing cultural nuances and stylistic challenges through collaborative reasoning. This approach not only improves translation quality but also extends LLM applications to linguistically and culturally rich domains. These contributions underscore the importance of tool use and agent collaboration in advancing translation education (Zheng et al., 2024).

A.2.4 Storytelling

Storytelling applications leverage LLM agents to create immersive and interactive learning experi-

ences (Simon and Muise, 2022). STARie (Li and Xu, 2023), a peer-like embodied conversational agent, integrates multimodal tools such as speech synthesis and facial animation to scaffold children's storytelling, fostering narrative creativity and oral communication skills (Beredo and Ong, 2021; Cassell, 2022). StoryAgent (Sohn et al., 2024) combines top-down story drafting with bottom-up asset generation to transform simple prompts into coherent, multi-modal digital narratives. By automating complex storytelling workflows (Liem et al., 2023), it democratizes content creation and enhances engagement in language learning. LLaMS (Zhang et al., 2025b), a multi-modal agent framework, is designed to generate multi-modal human-level stories characterized by expressiveness and consistency, incorporating the Story-Adapter module for long image sequence illustration. These systems demonstrate the potential of LLM agents to support both cognitive and creative aspects of language education for children.

A.2.5 Speaking

LLM agents are revolutionizing spoken language education by integrating reasoning and multi-agent collaboration to build adaptive dialogue systems (Liu et al., 2024d; Balan et al., 2024). ELLMA-T (Pan et al., 2024) employs contextual reasoning and role-playing in social VR environments to provide personalized feedback and language assessments, enabling learners to practice speaking in realistic scenarios (Lim et al., 2024; Li et al., 2025b). SpokenWOZ (Si et al., 2023) introduces a large-scale benchmark for task-oriented spoken dialogue, highlighting the importance of reasoning and multi-turn interaction in addressing real-world conversational challenges. FurChat (Cherakara et al., 2023), an embodied conversational agent, combines verbal and non-verbal communication cues to simulate natural interactions, making it a valuable tool for improving speaking skills through immersive and realistic practice. By employing multimodal signals such as speech and gestures, SpeechAgents (Zhang et al., 2024a) enhances the authenticity of dialogue simulations, capturing consistent content, natural rhythm, and rich emotional expression. Through Multi-Agent Tuning (Liang et al., 2024), it optimizes LLM capabilities for large-scale simulations involving up to 25 agents, enabling applications like drama creation and audio novel generation.

A.3 Agent for Professional Development

Agents for professional development harness the capabilities of LLMs to offer scalable, adaptive, and context-aware learning experiences tailored to domain-specific needs. This section summarizes how recent studies develop agents to revolutionize professional training in fields including *medical* (§A.3.1), *computer science* (§A.3.2), and *law education* (§A.3.3).

A.3.1 Medical Education

The deployment of LLM agents in healthcare has created new opportunities for personalized, interactive, and scalable systems, with several health agents introduced (Shusterman et al., 2025) such as Talk2Care (Yang et al., 2024c) and openCHA (Abbasian et al., 2023). Additionally, Abd-Alrazaq et al. (2023) highlight the educational potentials of LLMs in crafting personalized curricula, adaptive learning plans, and dynamic assessment tools for medical education, while concurrently addressing challenges including algorithmic bias, misinformation, and privacy issues. MEDCO (Wei et al., 2024), a multi-agent system, has the capacity to replicate real-world medical training environments through agent collaboration with virtual patients, expert physicians, and radiologists, enhancing interdisciplinary learning and peer interaction. Furthermore, Abbasian et al. (2023) introduce openCHA as a personalized LLM-powered framework that integrates external resources and orchestrates multi-step problem-solving for complex healthcare queries (Ye et al., 2024a), emphasizing tool use and action planning. Beyond traditional education, Steenstra et al. (2024) explore LLMs in creating fantasy narrative games for adolescent health education, demonstrating the agents’ ability to generate engaging, doctor-validated content that enhances knowledge retention through gamification. Li et al. (2024b) present Agent Hospital, a simulation environment where LLM-driven agents evolve through autonomous interactions, demonstrating significant improvements in medical reasoning and performance on benchmarks like MedQA (Jin et al., 2021) after treating thousands of simulated patients. Collectively, these investigations highlight the versatility of LLM agents within medical education, demonstrating their abilities in reasoning, collaboration, tool integration, and adaptive learning to effectively address a broad spectrum of educational and clinical challenges (Karabacak and Margetis, 2023; Tian et al., 2024b; Ullah et al., 2024).

A.3.2 Computer Science Education

An agent for computer science (CS) education greatly enhances learning by providing personalized guidance on coding, debugging, and understanding CS principles (Ma et al., 2024b; Lee and Song, 2024; Kosar et al., 2024; Liu et al., 2024b). For example, CodeAgent (Zhang et al., 2024b) serves as an LLM agent framework for repo-level code generation, incorporating external tools such as WebSearch and DocSearch. Recent studies have demonstrated the potential of agent-based code generation systems such as ToolCoder (Ding et al., 2025), SWE-agent (Yang et al., 2025), AgentCoder (Huang et al., 2023a), and MapCoder (Islam et al., 2024), which can significantly enhance students’ coding efficiency (Jin et al., 2024a; Wang et al., 2025a; Frankford et al., 2024). Furthermore, Bassner et al. (2024) introduce Iris, an LLM-driven virtual tutor designed to offer personalized, context-aware assistance to CS students within the interactive learning platform Artemis. Besides, Chen et al. (2024a) propose Learning-by-Teaching (LBT) as an effective pedagogical strategy for CS education, and leverage the advantages of LLM agents (*e.g.*, contextual conversation & learning from demonstrations).

A.3.3 Law Education

LLM agents leverage pre-trained legal knowledge, interactive capabilities, and reasoning skills to support law education through judicial interpretation, moot court simulation, and case analysis (Chen et al., 2024b; Nelson, 2024; Lai et al., 2024; Yuan et al., 2024). However, evaluations from LawBench (Fei et al., 2023) and LegalBench (Guha et al., 2023) reveal that LLMs struggle with legal knowledge application and judicial aid. LegalAgentBench (Li et al., 2024a) further highlights their limitations in multi-hop reasoning and defense statement writing, showing that LLM agents require significant improvements to effectively assist in complex legal tasks. Despite these challenges, LLM agents are emerging as valuable tools for moot court simulations, a crucial component of legal reasoning and advocacy training. DeliLaw (Xie et al., 2024a) enhances law education by integrating legal and case retrieval modules, enabling students to practice legal research, case analysis, statutory interpretation, and mock consultations. LawLuo (Sun et al., 2024) applies a multi-agent framework with retrieval-augmented generation to simulate multi-turn legal consultations, improving

personalization and ambiguity handling. Similarly, AgentCourt (Chen et al., 2024b) and AgentsCourt (He et al., 2024b) simulate courtroom interactions and judicial decision-making, providing a realistic training ground for law students. AgentsBench (Jiang and Yang, 2024) extends this by offering multi-agent legal reasoning and case analysis, further advancing AI-driven legal education.

B Datasets & Benchmarks

In Table 2, we provide a comprehensive summary of publicly available datasets and benchmarks designed to evaluate LLM agents for education across various domains. It categorizes resources based on their primary goal, target users, subject domain, education level, language, modality and dataset size. We hope this collection can support and advance research on LLM agents for education.

Several datasets are designed to evaluate the pedagogical agents, such as ASSIST09 (Feng et al., 2009) and Junyi (Chang et al., 2015), which support knowledge tracing (KT) in K-12 math education, while others like EduAgent (Xu et al., 2024a) facilitate adaptive learning (AL) by dynamically adjusting content based on student profiles. In addition, error correction and detection (ECD) datasets, such as Virtual Teacher (Xu et al., 2024b) and MathCCS (Zhang et al., 2025d), assess LLM agents’ ability to identify and rectify student mistakes in math learning. Other datasets cater to writing, reading, and language learning, including FABRIC (Han et al., 2023), EssayJudge (Su et al., 2025), and EX-CEGE (Ye et al., 2024b), which focus on feedback and generation (FCG) for student essays. Multi-Sim (Ryan et al., 2023) and Wang et al. (2024b) provide multi-lingual translation and storytelling benchmarks, expanding LLM capabilities beyond English-language education.

Several datasets support domain-specific educational agents across science, law, medicine, and computer science. Beyond their primary goal of evaluating pedagogical ability, these datasets assess LLM agents in domain-specific applications. They provide insights into how LLM agents can be adapted for specialized instruction, evaluating their ability to deliver subject-specific knowledge, facilitate problem-solving, and enhance interactive learning experiences across diverse educational fields. ScienceAgentBench (Chen et al., 2024d) and TheoremExplainBench (Ku et al., 2025) assess scientific reasoning and theorem explanation,

while ML-Bench (Tang et al., 2023) and MLAgentBench (Huang et al., 2023c) focus on machine learning education. In law education, datasets like LawBench (Fei et al., 2023), LegalBench (Guha et al., 2023), and AgentCourt (Chen et al., 2024b) evaluate legal knowledge application, case analysis, and court simulations. Medical education datasets, including MedBench (Cai et al., 2024) and OmniMedVQA (Hu et al., 2024), test clinical reasoning and medical knowledge retrieval. For computer science, SWE-Bench (Yang et al., 2025) and Programming Feedback (Estévez-Ayres et al., 2024) assess code generation, debugging, and software engineering instruction. These benchmarks help refine LLM agents for specialized tutoring, enhancing AI-driven learning in professional fields.

Table 2: Summary of existing datasets and benchmarks of LLM agents for education.

Dataset&Benchmark	Goal	User	Domain	Level	Language	Modality	Amount	Source
ASSIST09	KT	Student	Math	K12	EN	text	227k	(Feng et al., 2009)
Junyi	KT	Student	Math	K12	ZH	text	2.5M	(Chang et al., 2015)
EduAgent	AL & CS	Student	-	Graduate	EN	text & image	1,015	(Xu et al., 2024a)
MathDial	AL & KT	Student	Math	K12	EN	text	45	(Macina et al., 2023)
MultiArith	AL	Student	Math	K12	EN	text	180	(Xu et al., 2024a)
CoMTA	KT	Student	Math	K12	EN	text	153	(Scarlato et al., 2025)
MaCKT	KT	Student	Math	K12	EN	text	452	(Yang et al., 2024b)
Virtual Teacher	ECD	Student	Math	K12	ZH	text & image	420	(Xu et al., 2024b)
MathCCS	ECD	Student	Math	K12	ZH	text & image	420	(Zhang et al., 2025d)
MathTutorBench	ECD & FCG	Student	Math	K12	EN	text	7 tasks	(Macina et al., 2025)
MultiSim	-	Student	Reading	-	Multi-lingual	text	1.7M	(Ryan et al., 2023)
FABRIC	FCG	Student	Writing	-	EN	text	1,782	(Han et al., 2023)
EssayJudge	FCG	Student	Writing	-	EN	text & image	1,054	(Su et al., 2025)
PROF	FCG	Teacher	Writing	-	EN	text	363	(Nair et al., 2024)
EXCGEC	FCG	Student	Writing	-	ZH	text	8,216	(Ye et al., 2024b)
Wang et al. (2024b)	-	All	Translation	-	Multi-lingual	text	70K	(Wang et al., 2024b)
NewEpisode	-	Student	Storytelling	-	EN	text & image	24.5K	(Wang et al., 2024e)
SD-Eval	ECD	Student	Speaking	-	EN	speech	7,303	(Ao et al., 2024)
Programming Feedback	FCG	Teacher	Computer Science	-	Code	text	52	(Estévez-Ayres et al., 2024)
Review Critique	FCG	All	Computer Science	-	EN	text	440	(Du et al., 2024)
AAAR-1.0	FCG	All	Computer Science	-	EN	text & image	1,000	(Lou et al., 2024)
ScienceAgentBench	-	All	General Science	-	EN	text	102	(Chen et al., 2024d)
TheoremExplainBench	-	All	General Science	-	EN	text	240	(Ku et al., 2025)
MLGym-Bench	-	All	General Science	-	EN	text	13 tasks	(Nathani et al., 2025)
ML-Bench	-	All	Machine Learning	-	EN	text & image	9,641	(Tang et al., 2023)
MLAgentBench	-	All	Machine Learning	-	EN	text	13 tasks	(Huang et al., 2023c)
SciCode	-	All	General Science	-	EN	text	338	(Tian et al., 2024a)
BLADE	-	All	General Science	-	EN	text	12 tasks	(Gu et al., 2024)
DiscoveryBench	-	All	General Science	-	EN	text	1,167	(Majumder et al., 2024)
SUPER	-	All	General Science	-	EN	text	796	(Bogin et al., 2024)
E-EVAL	-	All	Math & Language & General Science	K12	ZH	text	4,351	(Hou et al., 2024)
MedBench	-	All	Medical	-	ZH	text	40,041	(Cai et al., 2024)
CMB	-	All	Medical	-	ZH	text	280,839	(Wang et al., 2024d)
OmniMedVQA	-	All	Medical	-	ZH	text & image	127,995	(Hu et al., 2024)
MedEval	-	All	Medical	-	EN	text & image	22,779	(He et al., 2023)
OSWorld	-	All	Computer Science	-	EN	text & image	369	(Xie et al., 2024b)
Spider2-V	-	All	Computer Science	-	EN	text & image	812	(Cao et al., 2024)
VisualWebArena	-	All	Computer Science	-	EN	text & image	910	(Koh et al., 2024)
WebArena	-	All	Computer Science	-	EN	text & image	812	(Zhou et al., 2023)
SWE-Bench	-	All	Computer Science	-	EN	text	2,294	(Yang et al., 2025)
SWE-Bench M	-	All	Computer Science	-	EN	text & image	617	(Yang et al., 2024a)
Magentic-One	-	All	Computer Science	-	EN	text & image	617	(Fourney et al., 2024)
Lawbench	-	All	Law	-	ZH	text	20 tasks	(Fei et al., 2023)
LegalBench	-	All	Law	-	EN	text	162 tasks	(Guha et al., 2023)
LegalAgentBench	-	All	Law	-	ZH	text	300 tasks	(Li et al., 2024a)
Agentcourt	-	All	Law	-	ZH	text	550	(Chen et al., 2024b)
SimuCourt	-	All	Law	-	ZH	text	420	(He et al., 2024b)